Project 2 Proposal: Supply Chain Network Optimizer

Dharmesh Sharma *Mit Gandhi*

Project Overview

Let's consider what occurs when one makes an online purchase. Before that package reaches your door, it passes through a chain of steps: first, the item is manufactured by the supplier; then it goes into storage in a warehouse; later it gets sorted at the distribution center; finally, it gets delivered via the truck to the consumer.
Companies running these supply chains are constantly trying to answer two questions:

- What order do things need to happen in?
- What is the cheapest or fastest way to get goods from point A to point B?

Our project takes a real supply chain dataset and uses graph algorithms to answer both of those questions through an interactive web page, and it is easy enough to use that anyone, regardless of technical background, can open it and immediately understand what's going on.

Dataset

We are using the **DataCo Smart Supply Chain** dataset, freely available on Kaggle (by DataCo Global). It is one of the most widely referenced supply chain datasets and it comes with a dedicated column description file, making it easy to validate and work with.
Key facts:

- 180,000+ order records
- 50+ features per record
- Global coverage spanning multiple countries and regions
- Product types include clothing, sports equipment, and electronics
- Covers provisioning, production, sales, and commercial distribution

Algorithm and Data Structure

The core data structure is a **directed weighted graph** where nodes are supply chain stages and edges are shipping routes weighted by cost or delivery time.
Topological Sort (Kahn's Algorithm):

- Treats the supply chain as a directed acyclic graph (DAG) since goods only flow forward: supplier → warehouse → distribution center → store
- Produces a valid processing order, ensuring no stage is scheduled before the one it depends on
- The output determines how the graph is laid out visually in the app

Dijkstra's Shortest Path:

- Finds the minimum-cost or fastest route between any two points in the network
- Uses real shipping costs and delivery times from the DataCo dataset as edge weights
- Runs every time the user selects a new origin and destination
- Returns the full step-by-step path with total cost and estimated delivery time

The two algorithms work together, where topological sort defines the valid structure of the chain, and Dijkstra's finds the best way to move through it.

Application Functionality

The web page will be built using Streamlit with three main sections:

1. Supply Chain Graph View: It will be an interactive visual graph where nodes represent supply chain stages and arrows show the flow of goods, laid out in the correct order by the topological sort.

2. Route Optimizer: In this, the user picks an origin, a destination, and whether to optimize for cost or speed, then hits a button to run Dijkstra's and see the cheapest or fastest path highlighted on the graph with a cost and delivery time breakdown.

3. Dataset Explorer: It provides summary charts from the DataCo data showing shipping costs by region, on-time vs. late deliveries, and most frequently used routes.

Expected Outcome

By the end of the project, we aim to have a fully working web app that lets any user explore a real supply chain network and find the best route between any two points in it. The key outcomes are:

- A correctly implemented topological sort that structures and displays the supply chain graph
- A working Dijkstra's implementation that returns optimal routes based on real dataset values
- An interactive Streamlit app accessible to a non-CS audience

The final project will include a GitHub repository containing the source code, setup instructions, and documentation, along with a short video demonstration explaining how the application works.

AI Usage:

Google – Dataset search

Youtube – Different project ideas for Graph vs Dynamic vs NP-problems.

Claude – To check if we can implement topological sort with Dijkstra's Algorithm together.